A Computer Vision Framework Combining Machine Learning and Deep Learning for Multimodal Triage of Stroke, Hemorrhage, and Mass Effect on Brain CT and MRI

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Abstract

Acute stroke, intracranial hemorrhage, and traumatic brain injury require prompt assessment because delays in recognizing critical findings can affect treatment decisions and patient outcomes. However, timely interpretation of brain imaging may be difficult in resource-constrained settings where radiology expertise, specialist coverage, and diagnostic infrastructure are limited. This article presents the design of a computer-vision triage framework for brain computed tomography (CT) and magnetic resonance imaging (MRI), bringing data science and analytics together with clinical medicine and surgery. Its objective is to support the prioritization of stroke, hemorrhage, mass effect, and apparently normal studies without replacing professional diagnosis. The workflow accepts CT images, MRI images, or both CT and MRI images within a single study, while retaining the modality assigned to each image. It applies documented image validation, modality-aware preprocessing, and quantitative feature extraction, and defines a comparative modeling plan for feature-based machine learning and convolutional-neural-network deep learning. The application is designed to combine image-derived screening impressions with clinician-entered presenting history and examination information, then present structured findings, confidence, priority, and differential considerations for qualified review. Evaluation is specified using patient-level data partitions and clinically relevant measures of discrimination, sensitivity, specificity, F1-score, area under the receiver-operating-characteristic curve, calibration, subgroup performance, and robustness across modalities, sites, scanners, and acquisition conditions. The framework also incorporates explainability, privacy safeguards, data governance, human oversight, and implementation considerations for African and other settings with constrained resources. The current application is a research prototype, and no clinical performance or outcome benefit is claimed. Such claims require representative, appropriately annotated datasets, external validation, and prospective workflow evaluation. This work therefore contributes a reproducible design and validation pathway for responsible AI-assisted neuroimaging triage.

Keywords: brain imaging; computed tomography; magnetic resonance imaging; stroke; hemorrhage; mass effect; deep learning; explainable AI; triage; clinical decision support

1 Introduction

Stroke is a leading cause of death and disability worldwide, and the clinical value of treatment is strongly time dependent (WHO 2022; Feigin et al. 2022; Campbell and Khatri 2020). Emergency neuroimaging is therefore not merely a diagnostic activity; it is also a prioritization mechanism. Non-contrast CT is widely available and fast, while MRI supplies complementary information about tissue injury, diffusion restriction, perfusion, and lesion age (Powers et al. 2019; Wintermark et al. 2013; Chalela et al. 2007). Intracranial hemorrhage and mass effect represent particularly important urgent findings because they may require immediate neurosurgical, neurocritical-care, or blood-pressure management decisions (Greenberg et al. 2022; Hemphill et al. 2001).

Radiology workloads, unequal access to subspecialty expertise, and the increasing volume of imaging create a practical need for computer-assisted triage. A triage model does not need to make the final diagnosis to be useful: it may identify studies that should be reviewed sooner, summarize image-derived evidence, and make the reasoning pathway auditable. However, clinical deployment requires careful separation of screening from diagnosis, explicit uncertainty, robust external validation, and human oversight (Topol 2019; Kelly et al. 2019; Holzinger et al. 2019).

This article develops a combined CT–MRI research and software framework. Its contributions are:

1. A single workflow that retains modality identity for every uploaded image.
2. A comparison between feature-based machine learning and image-based deep learning.
3. Clinician-oriented outputs for stroke, hemorrhage, mass effect, and normal categories.
4. A reproducible evaluation and governance plan.

2 Clinical and Research Background: Literature Review

2.1 Clinical context and imaging roles

Ischemic stroke results from interrupted cerebral perfusion, while hemorrhagic stroke involves bleeding within or around the brain. Traumatic brain injury can cause hemorrhage, edema, contusion, and secondary mass effect. Mass effect describes displacement or compression of intracranial structures and may arise from several underlying conditions. These findings may coexist; therefore, a four-class label is a useful prototype simplification, not a complete representation of clinical diagnosis.

Non-contrast CT is fast and widely available in emergency care and is central to identifying acute intracranial blood. MRI, including diffusion-weighted sequences, can detect early ischemic tissue injury and provide complementary characterization, but acquisition time, scanner access, and patient stability affect its use (Chalela et al. 2007; Warach et al. 1992; Kidwell et al. 2004; Wintermark et al. 2013). The clinical implication is that CT and MRI should be treated as complementary, modality-specific inputs. A model must not infer that the absence of a finding on one modality guarantees its absence on the other.

2.2 Literature review approach

This narrative review synthesizes clinical imaging guidance, representative neuroimaging-AI studies, foundational computer-vision methods, and work on explainability and clinical implementation. The cited works are used to identify methods and research gaps; this is not a registered systematic review or a quantitative meta-analysis. Study selection for a final dissertation should be documented with search dates, databases, exact query strings, inclusion criteria, and screening counts. Google Scholar and ResearchGate are useful discovery and access services, but study metadata and claims should be verified against the publisher record or bibliographic database before submission.

2.3 Representative prior work

Earlier research demonstrates that deep-learning systems can detect selected urgent findings in head CT, while broader medical-imaging reviews describe the strengths and limitations of convolutional models (Titano et al. 2018; Chilamkurthy et al. 2018; Kuo et al. 2019; Arbabshirani et al. 2018; Litjens et al. 2017; Shen et al. 2017). These studies establish feasibility for defined tasks; they do not, by themselves, validate the present four-label, CT–MRI, mixed-study prototype or demonstrate benefit in the intended local setting. Table 1 summarizes the relevance and remaining gap without treating results from different datasets as directly comparable.

Table 1: Representative prior work informing the proposed study. The table summarizes study focus, not comparative performance; datasets and endpoints differ across publications.

Study	Imaging/task focus	Contribution relevant to this thesis	Limitation or remaining question addressed here
(Chalela et al. 2007)	Emergency stroke imaging; CT and MRI	Compared the role of CT and MRI in acute stroke assessment.	Clinical comparison does not supply a validated multimodal AI triage pipeline.
(Chilamkurthy et al. 2018)	Critical findings on head CT	Demonstrated deep-learning analysis for several urgent CT findings.	Focus is CT; local transferability and the present label set require separate evaluation.
(Titano et al. 2018)	Cranial imaging surveillance	Investigated automated prioritization of acute neurological events.	Workflow and dataset context differ; prospective benefit cannot be assumed for this prototype.
(Kuo et al. 2019)	Intracranial hemorrhage on head CT	Evaluated deep-learning support for a focused hemorrhage-detection task.	Single-finding CT performance does not establish four-class, cross-modality performance.
(Arbabshirani et al. 2018)	Intracranial hemorrhage on brain CT	Examined machine learning for automated CT identification.	External validation, class definitions, and workflow outcomes remain context dependent.
(Litjens et al. 2017; Shen et al. 2017)	Reviews of deep learning in medical imaging	Summarized architectures, applications, and recurring data challenges.	Reviews motivate a task-specific protocol but do not validate the current software or its outputs.

2.4 Evidence synthesis and research gap

Figure 1 presents a qualitative synthesis of the literature. It is a conceptual evidence map, not a numerical graph of publication counts or model accuracy. The map shows how the present thesis connects established, task-specific imaging-AI work to an explicit multimodal prototype and a validation plan.

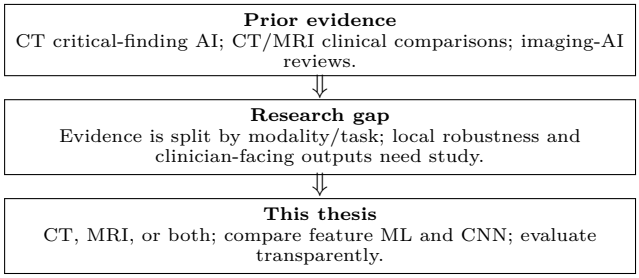


Figure 1: Qualitative literature-to-thesis evidence map. Arrows indicate research motivation, not proven causal or performance improvement.

2.5 Why this thesis is important

The thesis is important for three connected reasons. Clinically, it focuses on prioritizing time-sensitive brain findings while keeping a professional responsible for final interpretation. Methodologically, it compares interpretable engineered features with image-based deep-learning approaches and requires patient-level and external evaluation rather than relying on accuracy alone. From a data-science and health-systems perspective, it makes modality identity, data quality, uncertainty, governance, and resource constraints explicit design variables. The study's contribution is a reproducible evaluation pathway and a testable prototype, not a claim that the current system is ready for diagnosis or has already improved outcomes.

3 Problem Statement and Objectives

3.1 Problem statement

Many existing studies address a single modality or a single finding. A practical emergency tool must handle heterogeneous studies, preserve CT/MRI provenance, and communicate its output in language that supports review. The problem addressed here is the design of an auditable, multimodal, multi-class triage system for stroke, hemorrhage, mass effect, and normal studies.

3.2 General objective

To develop and evaluate a multimodal clinical decision-support prototype that analyzes brain CT and MRI images and assigns a clinically interpretable triage category for stroke, hemorrhage, mass effect, and normal findings.

3.3 Specific objectives

1. To design a data pipeline that preserves CT and MRI modality identity throughout ingestion, preprocessing, modeling, and reporting.
2. To compare feature-based machine-learning models with image-based deep-learning models.
3. To evaluate sensitivity, specificity, precision, F1-score, AUROC, calibration, and workload implications.
4. To develop clinician-oriented outputs using feature attribution and image-saliency explanations.
5. To assess model stability across sites, scanners, protocols, and image-quality conditions.
6. To define reproducible governance, validation, monitoring, and human-oversight requirements for safe use.

3.4 Research questions

The research questions are:

1. How does performance differ between CT and MRI?
2. How do radiomic and deep representations compare?
3. Which findings are most reliably prioritized?
4. How stable are predictions under site and scanner variation?
5. Can a concise result card improve the speed and clarity of clinical review without encouraging automation bias?

4 Materials and Methods

4.1 Study design

This is a retrospective, multi-stage development study followed by external and prospective evaluation. The development phase is intended to use de-identified images organized into training, validation, and held-out test partitions. Patient-level splitting is mandatory to avoid leakage between slices or studies from the same patient. A proposed initial partition is 70% training, 15% validation, and 15% test, subject to feasibility, site representation, and sufficient class counts. The final test set must remain inaccessible until model selection and threshold specification are complete. If the available dataset is too small for a stable three-way split, grouped cross-validation should be used during development and an independent external cohort reserved for final evaluation.

4.2 Data sources and labeling

Candidate data may be obtained from institutionally approved archives, public research datasets, and controlled-access repositories. No dataset size or class distribution is asserted in this article because a curated, labeled study cohort has not yet been documented in the workspace. Each case should have a de-identified patient and study identifier, modality and sequence, acquisition metadata where available, label provenance, and quality-control status. Labels should be defined in a written annotation manual, assigned by qualified readers, and adjudicated for disagreement. Weak labels derived from reports must be identified separately from expert annotations. Data inclusion and exclusion counts, missingness, duplicate handling, and reasons for exclusion must be reported in the final results.

The four prototype labels are:

- *stroke*
- *hemorrhage*
- *mass effect*
- *normal*

Multi-label cases should be retained in the source registry even if the initial classifier uses a mutually exclusive output. A future multilabel model may represent coexisting abnormalities more faithfully.

4.3 Preprocessing

The preprocessing pipeline standardizes orientation and image size, converts supported formats to a controlled representation, normalizes intensities, and records every transformation. CT preprocessing may include a documented windowing policy for brain and blood-sensitive views. MRI preprocessing may include bias-field correction, resampling, intensity standardization, and sequence harmonization when metadata are available. No transformation may use information from the held-out test set.

Images with severe motion, missing slices, corrupted pixels, or incompatible metadata are flagged rather than silently discarded. The software records the original path, transformed path, modality, dimensions, and quality-control decision. Data augmentation is applied only to training data and may include small rotations, translations, intensity changes, and noise consistent with clinical acquisition.

4.4 Feature engineering

The feature-based branch computes intensity statistics, entropy, texture descriptors, gradient or edge density, local contrast, and shape or region summaries when segmentation is available. Let x denote the feature vector and y the clinical class. A standardized feature matrix is formed as

$$z_j = \frac{x_j - \mu_j}{\sigma_j + \epsilon}, \tag{1}$$

where the mean and standard deviation are fitted on the training partition only. Feature engineering is evaluated for stability because a feature that changes substantially across scanners may create a misleading shortcut.

4.5 Machine-learning models

The classical branch compares logistic regression, support-vector machines, random forests, and gradient-boosted trees. Hyperparameters are selected using nested cross-validation on the training data. Class weighting or balanced sampling is considered when prevalence is unequal. The final model is serialized together with its feature schema, preprocessing parameters, training data version, and software environment.

4.6 Deep-learning models

The image branch evaluates transfer-learning backbones such as ResNet, DenseNet, and EfficientNet. A baseline model uses a frozen feature extractor followed by a task-specific head; a second configuration fine-tunes selected upper layers. The loss function may use weighted cross-entropy or focal loss when rare urgent findings are underrepresented. Early stopping is based on validation loss and clinically relevant metrics rather than training accuracy alone.

4.7 Proposed model structure and comparison

The planned comparison contains two alternatives, not an assumed fused clinical model. The feature-based branch receives the engineered feature vector and uses a standardized multiclass classifier. A transparent baseline is multinomial logistic regression trained with softmax cross-entropy; random forest and support-vector models may be added only if their dependencies and evaluation are reproducible. For input feature vector $x \in \mathbb{R}^p$, the class probability for class k is

$$P(y = k \mid x) = \frac{\exp(w_k^\top x + b_k)}{\sum_{j=1}^{K} \exp(w_j^\top x + b_j)}, \qquad K = 4. \tag{2}$$

The image-based branch receives a preprocessed image tensor and uses a 2-D convolutional backbone initialized from public pretrained weights when licensing and domain suitability permit. The backbone output is globally pooled and passed to a task-specific dense layer, dropout regularization, and a four-class softmax head. CT and MRI are routed with modality metadata and evaluated separately; a late-fusion model that combines examinations is exploratory and must not be reported as validated unless separately trained and tested on paired patient studies.

Figure 2 summarizes the planned comparative architecture. It shows data flow and candidate model components; it does not mean the CNN branch is already implemented or trained.

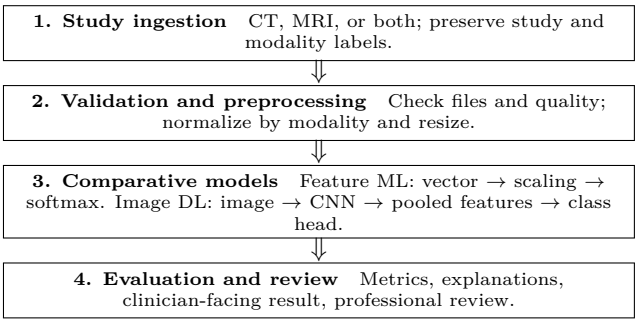


Figure 2: Proposed comparative model architecture. The CNN branch is planned research work; the current running image-triage path uses heuristic feature scores, as described in the prototype-status results.

Table 2 clarifies the inputs, structure, outputs, and current development status of each path.

5 Multimodal CT–MRI Workflow

The workflow accepts CT images, MRI images, or both CT and MRI images within a single study. Users may choose CT-only or MRI-only analysis, or upload CT and MRI files together in the combined mode. For combined uploads, the system creates one study list while preserving the modality label assigned to every image; the maximum is 40 images total. Each image then passes through modality-specific preprocessing, shared feature extraction, triage prediction, explanation, and result rendering.

The combined study summary reports filename, modality, screening impression, priority, and confidence. This design prevents an MRI image from being interpreted using CT assumptions and makes it possible to compare findings across modalities. When several images are submitted, the system presents per-image outputs and a study-level table rather than silently collapsing all evidence into one unsupported score.

A reference processing sequence is:

1. register the patient or study identifier;
2. select CT, MRI, or CT + MRI;
3. upload no more than 40 supported images;
4. validate file type, image readability, and modality assignment;
5. preprocess each image using the assigned modality;
6. compute features and run the selected model;
7. generate explanation and clinician-readable result;
8. export an auditable report for qualified review.

6 Explainability and Clinical Output

A prediction without a reason is difficult to audit. For the feature branch, explanations rank the features that contributed most strongly to the result. For the image branch, Grad-CAM or related saliency methods highlight image regions associated with the model output (Selvaraju et al. 2017). SHAP provides a unified attribution framework for many model families (Lundberg and Lee 2017), while LIME can support local surrogate explanations (Ribeiro et al. 2016).

Explanations must not be presented as proof of pathology. Saliency maps can be unstable, and highlighted regions may reflect acquisition artifacts. The user interface therefore labels them as model evidence and includes the original image, processed image, modality, confidence, and limitations. The prototype result uses four practical priorities: Critical, Urgent, Moderate, and Routine. These labels are workflow suggestions, not treatment instructions.

The clinician-facing output uses plain language: *screening impression*, *priority*, *confidence*, *clinical interpretation*, and *recommended next step*. It also states that the output is a screening aid and not a definitive diagnosis. Such wording is important for reducing automation bias and preserving the responsibility of the reviewing professional (Amann et al. 2020; Sendak et al. 2020).

Table 3 specifies the intended report fields. The diagnosis field is explicitly a screening impression. In the current prototype, the differential list is a fixed template selected from the predicted category; it is not generated from a validated clinical knowledge base and must be checked by a qualified clinician.

7 Evaluation Plan

The primary evaluation uses patient-level held-out data. Accuracy is reported with sensitivity, specificity, precision, macro-F1, weighted-F1, balanced accuracy, AUROC, area under the precision-recall curve, calibration error, and confusion matrices. For urgent classes, sensitivity and negative predictive value are especially important. Confidence intervals are estimated with bootstrap resampling, and model comparisons use paired patient-level analyses where appropriate.

A modality-stratified analysis reports CT-only, MRI-only, and combined-study performance. Additional strata include scanner manufacturer, institution, age group, sex where ethically justified, acquisition protocol, and image quality. External validation is performed

Table 2: Proposed model branches and current implementation status.

Branch	Input	Model structure	Status and required evidence
Prototype image triage	One normalized grayscale image plus CT/MRI label	Six summary image features, heuristic class scores, fixed priority mapping	Implemented as a software prototype; not learned from a labeled cohort and not clinically validated.
Feature-based ML	Eight engineered image features from dataset loader	Training-only scaling, multinomial softmax/logistic baseline; optional comparator models	Training code exists; correct split use, non-placeholder held-out metrics, calibration, and external testing are required before performance claims.
Image-based DL	2-D image tensor with modality metadata	Transfer-learning CNN, global average pooling, dropout, four-class output head	Proposed experiment; no CNN training or comparative result is claimed in this article.
Study-level fusion	Paired CT/MRI studies, if available	Optional late fusion of modality-specific outputs with missing-modality handling	Future work requiring paired cases and independent validation.

Table 3: Expected clinician-facing result structure. Bracketed values are placeholders, not observed clinical results.

Output field	Example format	Interpretation and safeguard
Study and image	De-identified study ID; image filename	Supports traceability; should not expose direct identifiers in research exports.
Modality	CT, MRI, or both (per-image modality retained)	Makes the acquisition context explicit.
Screening impression	Stroke / hemorrhage / mass effect / normal / indeterminate	Model-generated category for review, not a confirmed diagnosis.
Priority	Critical / Urgent / Moderate / Routine	Prototype workflow tier; thresholds require clinical validation.
Confidence or score	[To be populated after calibration]	Current heuristic score is not calibrated probability and must not be interpreted as one.
Clinical context	Chief complaint, presenting illness, relevant history, examinations	User-entered context; it is not currently incorporated into the image classifier.
Differential considerations	[Template suggestions for clinician review]	Current suggestions are category-based static text, not model-generated differential diagnosis.
Explanation and next step	Rationale, features, prompt for professional review	Aids review; final interpretation and management remain with a clinician.

on a site that was not used during training. Prospective silent evaluation precedes any workflow intervention.

8 Results and Discussion

8.1 Current prototype results

At the time of writing, the project workspace contains no configured data directory or documented, labeled CT/MRI cohort. Consequently, no training sample size, class prevalence, test-set predictions, sensitivity, specificity, AUROC, F1-score, calibration result, or clinical workflow outcome can be reported. This is a substantive result of the project status: the software interface and analysis components are a prototype, while empirical validation remains pending. Table 5 distinguishes implemented functions from results that require data.

The implementation review identifies two methodological constraints that must be resolved before reporting predictive performance. First, the current app's training button constructs features from all rows returned by the dataset scanner, which includes train, validation, and test folders, rather than selecting training rows only. Using these rows for fitting would contaminate held-out evaluation. Second, the training function currently returns placeholder accuracy and F1 values rather than metrics calculated from an independent test set. These values must not be interpreted or published as model results. The current image triage path also uses manually specified feature-score weights, and its confidence calculation is uncalibrated. Therefore, a result card can be demonstrated as a software output, but it cannot presently be presented as a validated diagnosis or calibrated risk estimate.

8.2 Expected output and interpretation

Once approved data and evaluation are available, the expected output is a per-image result card containing study/image identifier, modality, screening class, priority tier, model score, explanation, and the clinical history entered by the professional. For combined studies, a summary table will show the result for every CT and MRI image, preserve modality identity, and flag discordant outputs for review. Differential considerations should be displayed as prompts, not as autonomous diagnoses. The example schema in Table 3 is illustrative and intentionally contains placeholders rather than invented patient-level values.

8.3 Discussion of model design and literature

The literature indicates that deep-learning methods can support narrowly defined urgent-image tasks, while clinical imaging guidance recognizes complementary roles for CT and MRI (Chalela et al. 2007; Wintermark et al. 2013; Chilamkurthy et al. 2018; Titano et al. 2018; Kuo et al. 2019). The present thesis extends that motivation by specifying a single modality-preserving workflow and a direct comparison between engineered-feature models and image-based CNNs. The comparison is scientifically useful because the models have different data needs and failure modes: engineered features may be easier to inspect and fit on small cohorts, whereas CNNs can learn spatial representations but may require more examples, careful regularization, and stronger tests for site and scanner shift. These are hypotheses to test, not findings already established by this project.

The model structure is deliberately staged. A simple baseline establishes whether the dataset and labels carry a learnable signal before a more complex CNN is justified. An external site and per-modality analysis are essential: pooled accuracy could conceal a model that performs well on CT but poorly on MRI, or performs well only on one scanner. Likewise, an urgent-class false negative has different clinical consequences from a false positive. Reporting macro-averaged measures, class-wise sensitivity, calibration, confidence intervals, and failure cases is therefore more informative than a single accuracy value.

The prototype's present heuristic classifier is useful for exercising image loading, display, modality routing, result formatting, and report export. Its fixed score weights, static differential suggestions, and uncalibrated confidence are not learned clinical evidence. The system should not be used to diagnose, rule out disease, or determine treatment. Before clinical testing, the project must replace or explicitly benchmark these heuristics, correct data-split handling, calculate metrics only on patient-disjoint held-out data, and review outputs with clinicians. The research is important precisely because it makes these engineering and evidentiary gaps visible rather than presenting a software demonstration as clinical efficacy.

Table 4: Recommended evaluation dimensions for the CT–MRI triage prototype.

Dimension	Measure	Clinical purpose
Discrimination	Sensitivity, specificity, AUROC, PR-AUC	Identify urgent studies while quantifying false alarms
Calibration	Reliability curve, Brier score, expected calibration error	Determine whether confidence is meaningful
Robustness	Site, scanner, protocol, and quality strata	Detect distribution shift
Workflow	Time to review, alert burden, override rate	Assess operational usefulness
Safety	Missed urgent findings, subgroup gaps, incident review	Govern deployment risk

Table 5: Results available at the current development stage and evidence still required.

Area	Current observation from the project	Result required for a completed empirical paper
Image intake	Interface supports CT-only, MRI-only, or mixed CT/MRI upload, up to 40 images per batch.	File acceptance and modality routing tested across representative formats and metadata.
Prototype image path	Grayscale resizing/normalization, six image summary features, heuristic category scores, and a rule-based priority are implemented.	Reader-annotated test performance, failure analysis, and calibration; heuristic output must be evaluated as its own baseline.
Feature model	A NumPy softmax classifier and dataset feature extraction exist.	Train/validation/test separation, correction of placeholder metrics, held-out predictions, uncertainty intervals, and external validation.
Deep-learning branch	Architecture and comparison protocol are specified in this paper.	CNN implementation, documented training, model selection, held-out evaluation, and modality-stratified results.
Clinical report	Result card and PDF generation are implemented; clinical history is captured as user input.	Usability assessment, assessment of differential templates, privacy review, and prospective workflow evaluation.

8.4 Interpretation of future experimental results

When experiments are conducted, the feature-model and CNN comparison should be interpreted against prespecified hypotheses and confidence intervals. If a CNN performs better on one modality but not another, that pattern should motivate modality-specific analysis rather than a claim of universal superiority. If an engineered-feature model is competitive, its lower complexity may be relevant for resource-constrained deployment, provided calibration and robustness are acceptable. If performance is weak, that result remains valuable: it may reveal insufficient labels, class imbalance, poor external generalization, or a need for three-dimensional and multi-label modeling. No direction or magnitude of effect is assumed in advance.

9 Ethics, Safety, and Governance

All data processing must comply with applicable ethics review, privacy, data-protection, and information-security requirements. Images should be de-identified before development, and the project should maintain a data dictionary and access log. The patient identification field in the prototype is a workflow identifier, not permission to expose direct identifiers in model logs or exported reports.

The system should be deployed only with a named clinical owner, documented intended use, versioned models, monitoring thresholds, rollback procedures, and an incident-reporting pathway. Changes to preprocessing, labels, thresholds, or model weights require validation before release. A clinical professional remains responsible for interpreting the complete examination and patient context.

Potential hazards include false reassurance, alert fatigue, automation bias, demographic or site-specific performance gaps, corrupted files, cyber-security events, and overinterpretation of confidence. Mitigations include conservative thresholds, visible uncertainty, human confirmation, regular drift monitoring, representative validation, and a clear prohibition on using the prototype as a standalone diagnostic system.

10 Software Implementation

The prototype is implemented as a modular Python application. Configuration defines labels, image extensions, target dimensions, dataset directories, model paths, and report paths. Dataset utilities enumerate train, validation, and test folders. The image pipeline performs loading and normalization. The triage pipeline computes prototype scores. The Streamlit interface supports CT, MRI, and CT + MRI uploads, a maximum of 40 images, patient or study identification, model training, clinician-readable results, and PDF export.

Reproducibility requires a locked environment, deterministic seeds where possible, source control, a dataset manifest, and automated tests for image loading, modality routing, label mapping, and report generation. The system should fail visibly when an image cannot be read rather than substituting a blank array. Every exported report should include the model version, timestamp, modality, image name, result, confidence, and a clinical disclaimer.

11 Discussion

The central design choice is to keep the workflow multimodal without pretending that CT and MRI are interchangeable. This is clinically plausible, operationally transparent, and compatible with incremental development. It also enables future research into late fusion, feature fusion, sequence-aware MRI modeling, and study-level aggregation.

The principal scientific risk is dataset bias. Models trained on curated datasets can learn scanner, site, annotation, or compression patterns. A high score may therefore reflect the data collection process rather than pathology. The study addresses this risk through patient-level separation, external validation, error analysis, calibration, and explicit subgroup reporting.

Another risk is overclaiming clinical benefit from retrospective performance. The article therefore defines the current application as a research prototype and prioritization aid. Clinical impact requires prospective workflow studies, usability assessment, regulatory review where applicable, and evidence that the tool improves timeliness without increasing missed findings or unnecessary escalation.

12 Limitations

The proposed labels simplify complex neurological findings. Stroke, hemorrhage, and mass effect can coexist, and a mutually exclusive classifier may hide clinically important combinations. Image-level analysis may also ignore three-dimensional context, prior examinations, clinical history, and laboratory data. The prototype currently

uses lightweight summary features and should not be interpreted as equivalent to a validated clinical deep-learning system.

Reference availability and dataset licensing may constrain replication. Public datasets often differ in acquisition and annotation, while institutional datasets may not be shareable. The bibliography includes methodological and clinical foundations, but every cited paper must be verified against the final database search and the study's inclusion criteria before submission.

13 Conclusion

This article specifies a practical CT–MRI triage research platform for stroke, hemorrhage, and mass effect. It integrates modality-aware ingestion, preprocessing, feature-based and deep-learning models, explainability, clinician-readable output, and governance. The combined mode permits CT and MRI images to pass through the complete process in one study while preserving modality identity. The framework is ready for controlled dataset ingestion and empirical evaluation; it is not a substitute for clinical diagnosis. The next stage is to populate the train, validation, and test folders with ethically approved data, execute the prespecified evaluation, and report both performance and failure modes.

Data and Code Availability

The project prototype, configuration, notebooks, and application are maintained in the accompanying AI Medical Imaging workspace. Dataset access is subject to the license and ethics conditions of each source. No patient-identifying data should be committed to the repository.

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